

Reducing Diagnostic Errors in Liver Disease Using a Multi-Modal AI Clinical Decision Support System

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1. Problem Statement

It is challenging for doctors to accurately diagnose liver diseases due to inconsistent clinical information and the high workload in many healthcare settings. Early signs of liver problems are often subtle and can be overlooked, leading to delayed or incorrect diagnoses that adversely affect patient outcomes. This issue is particularly pronounced in rural clinics and underserved areas, where access to specialized hepatologists and radiologists is limited. Current AI diagnostic tools typically analyze either medical images or clinical data independently, but rarely both in combination. Such separation increases the risk of misdiagnosis and reduces diagnostic accuracy.

Therefore, a comprehensive diagnostic system that integrates both visual imaging and clinical data is essential to provide more reliable and actionable guidance for liver disease diagnosis. Addressing this challenge holds significant societal value by potentially improving early detection rates, reducing healthcare disparities, and enhancing patient prognosis on a broad scale.

2. Initial Research with LLM Support

To begin our research, we used ChatGPT to gain a foundational understanding of diagnostic challenges in liver disease and the potential of AI-based solutions. We formulated clear questions for ChatGPT, such as:

- What are the common causes of diagnostic errors in liver disease?
- How do AI-based systems improve diagnostic accuracy in medical imaging?
- What are the latest advancements in multi-modal AI for healthcare?

ChatGPT provided detailed insights, including how diagnostic errors often arise from human variability and how multi-modal AI systems can integrate imaging and clinical data to enhance diagnostic accuracy. However, we did not solely rely on ChatGPT. We critically evaluated its responses by cross-referencing them with credible sources (e.g., peer-reviewed articles, medical guidelines).

3. Literature Review

A growing body of research has focused on the use of AI in diagnosing and managing liver diseases, with many studies leveraging imaging techniques such as ultrasound, CT, and MRI to detect abnormalities like nonalcoholic fatty liver disease (NAFLD), liver fibrosis, and hepatocellular carcinoma (HCC) [1, 2, 3, 6]. Imaging-based AI systems, often employing convolutional neural networks (CNNs), demonstrate improved sensitivity, specificity, and reproducibility over traditional radiological methods by automatically extracting relevant features from scans [1, 2, 4]. Concurrently, AI models analyzing clinical and laboratory data through machine learning classifiers or regression methods provide superior risk stratification and outcome prediction compared to conventional clinical scoring systems [3, 5].

Despite these advancements, most AI applications are limited to single data modalities, which constrains their diagnostic power. A critical gap in current research is the underuse of multi-modal AI approaches that integrate imaging with clinical and molecular data such as genomics and proteomics, thereby offering a more holistic understanding of liver pathology. Multi-modal deep learning models that combine CT scans with pathology images have shown enhanced classification accuracy for liver cancer variants and metastases compared to unimodal models [7]. Similarly, models like TMI-CLNet, which fuse imaging, clinical, and

radiomic data, improve prognostic evaluation in chronic liver diseases, underscoring the potential of data fusion to capture complex disease patterns more effectively [8].

Nonetheless, challenges remain that limit AI's clinical adoption in hepatology. Imaging-based models face variability from differences in imaging protocols and equipment, which can reduce generalizability across healthcare settings [2, 6]. Clinical-data models may lack the morphological context necessary for precise disease characterization [3]. Furthermore, despite the invasiveness and cost, traditional diagnostics such as liver biopsy and serological tests remain the gold standards due to their accuracy and interpretability, which AI methods have yet to fully replicate [9, 10]. Additional barriers include concerns about algorithm transparency, data heterogeneity, and high computational demands, all of which slow the integration of AI tools into everyday clinical workflows.

In summary, while AI shows significant promise in both imaging and clinical data analysis independently, the integration of multi-modal data remains an underexplored but critical frontier. Expanding research in this area—including incorporating genomics and proteomics—has the potential to greatly improve diagnostic accuracy, enable personalized treatment plans, and ultimately reduce the burden of liver disease globally. Addressing these challenges will not only advance AI technologies but also contribute to equitable healthcare by empowering clinicians across diverse settings with better diagnostic tools.

4. Proposed Solution

To address diagnostic inaccuracies and delays in liver disease detection, we propose the development of a **Multi-Modal AI Clinical Decision Support System (CDSS)** that combines **machine learning**, **computer vision**, and **structured data analysis**. This system is designed to integrate and analyze both medical imaging (e.g., MRI, CT scans) and patient clinical data (e.g., lab results, patient history), providing clinicians with consistent, interpretable, and early diagnostic support.

4.1 AI and Data Science Technologies Used

Computer Vision (Deep Learning with CNNs)

- **Technology:** Convolutional Neural Networks (CNNs) will be used for image analysis. CNNs are especially well-suited for extracting spatial features and detecting visual anomalies in medical images, such as liver tumors or fibrosis.
- **Justification:** Liver abnormalities, especially in early stages, often appear as subtle patterns on CT/MRI scans that may be missed by human eyes. CNNs can be trained to recognize these patterns with high accuracy and consistency, reducing reliance on radiologist experience alone.

Machine Learning for Clinical Data

- **Technology:** Gradient Boosted Trees, Logistic Regression, or Neural Networks will be used to process structured clinical data such as liver enzyme levels, bilirubin, and patient risk factors (age, alcohol consumption, etc.).
- **Justification:** These models can detect patterns and relationships between variables that contribute to liver disease risk, improving prediction accuracy when imaging data alone is inconclusive.

Multimodal Learning and Feature Fusion

- **Technology:** A **multi-input neural network architecture** will be employed to combine outputs from both imaging (CNNs) and clinical data (dense layers or recurrent networks).
- **Justification:** Many liver conditions require both radiological and clinical insight. Multimodal models mimic the holistic approach a clinician would take, thus improving reliability and reducing diagnostic oversight.

Explainable AI (XAI)

- **Technology:** SHAP (SHapley Additive exPlanations), Grad-CAM, or LIME will be incorporated.
- **Justification:** Providing transparency is critical in clinical settings. These tools explain the AI's decision by highlighting influential features and regions in an image, thereby building clinician trust and ensuring safe adoption.

4.2 Positive Impact on Target Population

- **Clinicians:** Reduces diagnostic burden, minimizes human error, and provides explainable second opinions to support clinical decisions.
- **Patients:** Enables earlier detection of liver diseases, improving treatment outcomes and survival rates.
- **Healthcare Systems:** Improves diagnostic consistency across rural and urban areas, especially where access to hepatology experts is limited.

4.3 Why This Solution Is Suitable

- Early-stage symptoms are often subtle and easily missed by humans but can be detected through AI-driven pattern recognition.
- Multimodal integration mimics human diagnostic processes, offering a more comprehensive view of patient health.
- Scalability and adaptability allow deployment in both high-resource hospitals and under-resourced clinics via cloud or edge computing.

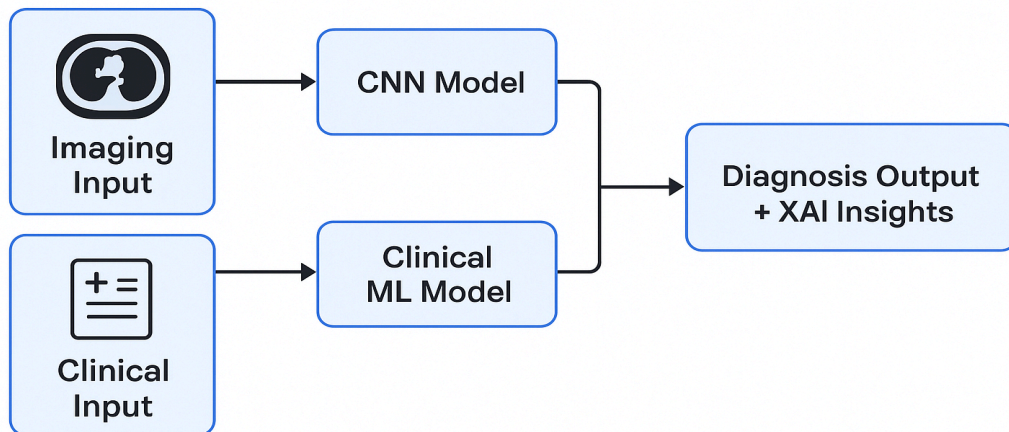


Fig 4.3.1 Workflow of Multi-Modal AI Clinical Decision Support System

5. Project Implementation Plan

The plan is structured into six realistic and sequential phases, each considering technical, resource, and time constraints.

Phase 1: Problem Scoping and Requirements Gathering

Objective: Define the scope, understand user needs, and determine technical requirements.

- **Activities:**
 - Conduct stakeholder interviews (e.g., clinicians, radiologists) to understand diagnostic workflows and pain points.
 - Document system goals, clinical requirements, data needs, and evaluation criteria.
- **Deliverables:** Functional and technical requirement specification.

Estimated Time: 1–2 weeks

Resources: Team consultation, clinical expert access

Phase 2: Data Collection and Management

Objective: Acquire and prepare imaging and clinical datasets for model development.

- **Data Sources:**
 - **Public Datasets:** Use established sources such as LiTS (Liver Tumor Segmentation), CHAOS, and MIMIC-IV for clinical and imaging data.

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- **Collaborative Data Sharing:** Partner with hospitals (pending ethics approval) to access de-identified real-world patient records.
- **Synthetic/Survey Data** (if needed): Simulated datasets or expert-generated samples for rare cases.
- **Activities:**
 - Data anonymization, cleansing, and formatting.
 - Merge and synchronize imaging and clinical datasets.
 - Perform data labeling and segmentation (semi-automated if possible).
- **Deliverables:** Integrated, preprocessed dataset with labeled inputs and outputs.

Estimated Time: 3–4 weeks

Resources: Public datasets, clinical collaborators, computing storage

Phase 3: Model Development and Training

Objective: Build and train the AI models for image analysis, clinical data processing, and multi-modal fusion.

- **Imaging Model:** Use CNN-based models (e.g., ResNet or U-Net) for liver segmentation and anomaly detection.
- **Clinical Data Model:** Use classifiers (e.g., Gradient Boosting, MLPs) to predict disease risk from structured health records.
- **Fusion Model:** Implement a hybrid neural network or attention-based fusion layer to integrate outputs from both modalities.
- **Activities:**
 - Hyperparameter tuning and cross-validation.
 - Implement explainability tools (SHAP, Grad-CAM).
 - Evaluate performance using metrics such as accuracy, precision, recall, and AUC.
- **Deliverables:** Trained models with documented performance metrics.

Estimated Time: 4–6 weeks

Resources: GPUs, AI developers, annotated datasets

Phase 4: User Interface and System Integration

Objective: Develop a functional prototype with a user-friendly interface for clinicians.

- **Design Considerations:**
 - Simple dashboard interface for entering patient data and uploading images.
 - Clear diagnostic output with explanation visualizations.
 - Compatibility with hospital IT systems (e.g., HL7, FHIR standards).
- **Activities:**
 - Create wireframes and conduct clinician usability tests.
 - Implement frontend (e.g., using React) and backend (e.g., Flask, FastAPI) integration.
 - Link AI predictions with display components and user inputs.
- **Deliverables:** Interactive UI prototype integrated with AI model.

Estimated Time: 3–4 weeks

Resources: UI/UX designers, frontend/backend developers

Phase 5: Validation and Pilot Testing

Objective: Test the system in a controlled clinical setting and refine it based on user feedback.

- **Activities:**
 - Conduct offline testing using previously unseen real-world patient cases to assess baseline model performance.
 - Deploy the system in a small-scale hospital or clinic for real-time use by clinicians.
 - Collect structured feedback from medical professionals through surveys, usability assessments, and interviews.
- **Evaluation Metrics:**
 - **Diagnostic accuracy:** Compare AI system outputs to diagnoses by expert clinicians using standard metrics such as accuracy, precision, recall, and F1-score.
 - Time required to reach a diagnosis and ease of system navigations using SUS and task-completion benchmarks.
 - User satisfaction and confidence in AI-generated outputs using Likert scales and open-ended feedback.

Deliverables: Pilot testing report and iteration plan.

Estimated Time: 3–5 weeks

Resources: Access to clinical test environment, ethics clearance, participating clinicians, and IT support team

Phase 6: Final Deployment and Monitoring

Objective: Launch the system in the real-world environment and ensure long-term success.

- **Activities:**
 - Full deployment on secure hospital servers or cloud platforms.
 - Monitor system accuracy and user engagement.
 - Implement continuous learning using clinician feedback and updated data.
 - Prepare documentation and training materials for users.
- **Deliverables:** Fully operational CDSS, user manual, ongoing maintenance plan.

Estimated Time: 2–3 weeks (deployment), continuous monitoring

Resources: IT team, DevOps, support staff

Feasibility Considerations

- **Time Constraints:** With phased implementation, a minimum viable product (MVP) can be developed within 4–6 months.
- **Resource Constraints:** By using publicly available datasets and open-source tools, development costs can be minimized.
- **Technical Risks:** Mitigated by modular development, thorough validation, and feedback integration.
- **Clinical Adoption:** Ensured through clinician co-design, usability testing, and transparent outputs.

6. Evaluation, Impact, and Ethics

Evaluation Strategy

To assess the success and effectiveness of the proposed Multi-Modal AI Clinical Decision Support System (CDSS), a combination of **quantitative model performance metrics**, **user-centric evaluations**, and **real-world impact measurements** will be applied:

1. Model Performance Metrics

- **Accuracy, Precision, Recall, and F1-Score:** These metrics will assess the system's ability to correctly identify liver disease cases, balancing both sensitivity and specificity.
- **Area Under the Curve – Receiver Operating Characteristic (AUC-ROC):** This will measure the model's ability to distinguish between healthy and diseased cases across various threshold settings.

- **Confusion Matrix Analysis:** This will provide detailed insight into diagnostic errors, such as false positives and false negatives, to identify patterns of misclassification.

2. Usability and User Feedback

- **Clinician Surveys and Interviews:** Structured feedback on ease of use, trust in AI suggestions, interpretability, and time saved.
- **System Usability Scale (SUS):** To quantitatively assess the usability of the interface and workflow integration.
- **Task Completion Time:** Time taken for clinicians to reach a diagnostic decision with and without the system.

3. Impact on the Target Community

- **Early Diagnosis Rates:** Measure any increase in detection of liver diseases at earlier stages.
- **Clinical Outcome Improvement:** Reduction in delayed or missed diagnoses through comparison with historical records.
- **Adoption in Underserved Areas:** Track uptake in rural or resource-limited clinics.

Ethical Considerations and Mitigation Strategies

The development of a healthcare AI system requires careful attention to ethical challenges to ensure fairness, transparency, and trust.

1. Privacy and Data Security

- **Risk:** The system may expose sensitive patient information during data processing, storage, or transmission.
- **Mitigation:**
 - All data will be de-identified prior to use, ensuring strict anonymisation protocols are followed.
 - Compliance with relevant data protection regulations such as HIPAA and GDPR.
 - Use of secure data storage and encrypted transmission protocols.

2. Fairness and Bias

- **Risk:** The model may underperform or produce skewed outcomes for certain demographic groups due to imbalanced training data (e.g., differences in ethnicity, gender, or geography).

- **Mitigation:**
 - Ensure diverse training datasets that reflect real-world populations.
 - Conduct fairness audits during model development and evaluation phases.
 - Regularly monitor predictions for equitable outcomes.

3. Transparency and Explainability

- **Risk:** Clinicians may mistrust or misinterpret AI decisions if they are opaque.
- **Mitigation:**
 - Integrate Explainable AI tools (e.g., SHAP, Grad-CAM) to provide visual and feature-based rationale for decisions.
 - Involve clinicians in model validation and refinement.

4. Accountability and Governance

- **Risk:** Misuse of the system or over-reliance on AI in critical decisions.
- **Mitigation:**
 - Establish clear guidelines that the system serves as a **decision support tool**, not a replacement for clinical judgment.
 - Audit logs and feedback loops will ensure traceability and support continuous model refinement.

7. Conclusion

This project proposes a **Multi-Modal AI Clinical Decision Support System** that combines **computer vision** and **machine learning** to enhance the diagnosis of liver diseases by integrating medical imaging and clinical data. The solution addresses the limitations of traditional diagnostic methods by improving accuracy, reducing clinician workload, and enabling early detection—especially in settings with limited access to specialists.

Through careful **model development**, **user-centered design**, and **ethical safeguards**, the proposed system can lead to measurable improvements in healthcare delivery. By empowering clinicians with actionable insights, particularly in underserved areas, this project has the potential to drive **positive social change**, **reduce health disparities**, and improve **patient outcomes** across diverse populations.

Contribution Sheet

Full Name	Student Number	Contribution Percentage	Signature
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